

02 — Deployment Architecture

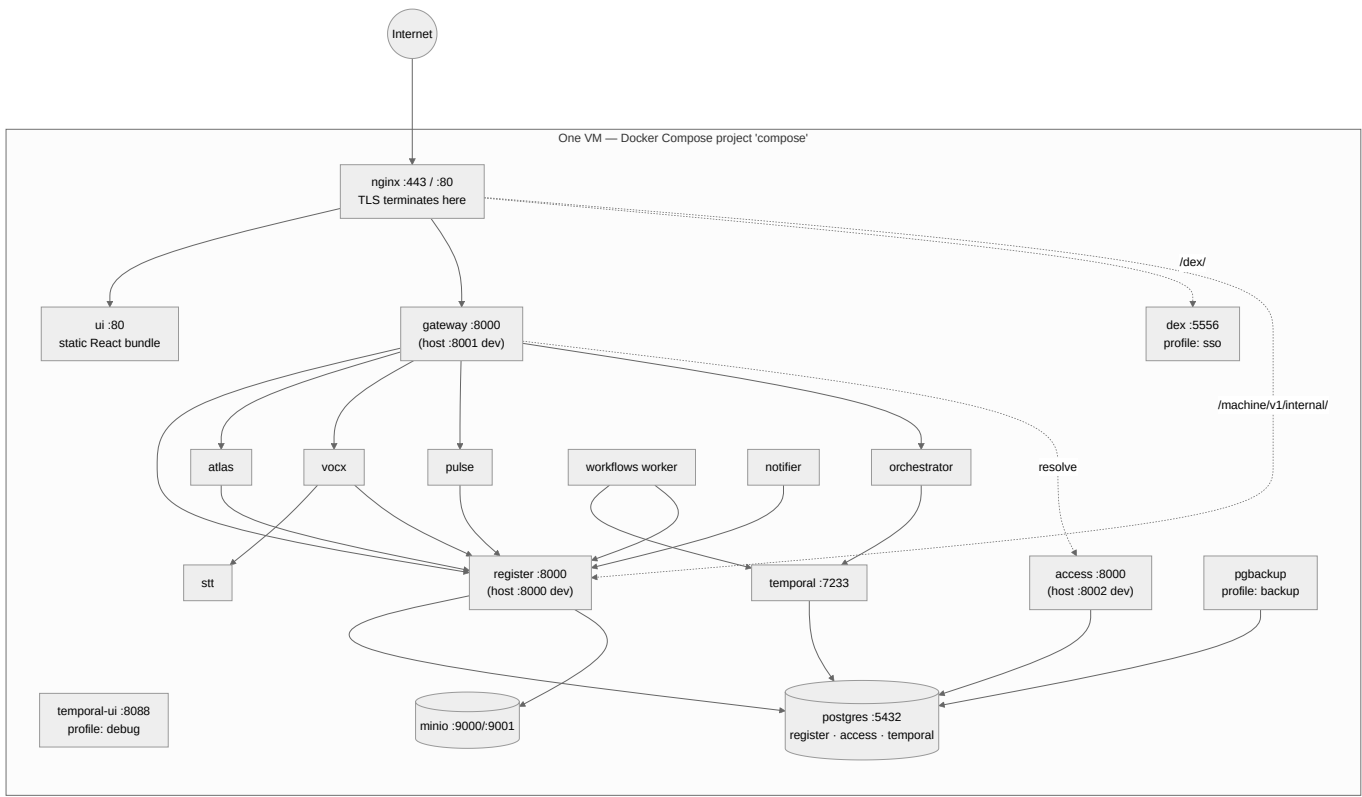
Audience: whoever deploys, operates or sizes PRISM. **Companion docs:** [10 Upgrade & rollback](#) · [09 Backup & restore](#) · [13 Operations](#) · [14 Configuration](#)

PRISM ships in two shapes from the same source tree:

Shape	Where it is used	Entry point
Docker Compose on one VM	Evam production (EC2), demos, laptops, WSL	deploy/compose/docker-compose.yml
Helm on Kubernetes	Larger / multi-replica deployments	deploy/helm/prism

Both build the *same images* from the *same Dockerfiles*. The difference is orchestration, not code.

1. Compose topology (production today)



The rule the diagram encodes

Only three things are reachable from outside: **nginx**, and **(in dev only) the debug ports**. **ATLAS**, **PULSE**, **VocX**, the **orchestrator**, the **UI** container and **STT** publish **no host port at all**. They are reachable only through the edge → **gateway**, and each accepts only the scoped service key the gateway injects. There is no way to call **ATLAS** around the authorization boundary.

Published ports

Port	Service	Present in production?
443 / 80	nginx (<code>EDGE_HTTPS_PORT</code> / <code>EDGE_HTTP_PORT</code>)	Yes — the only ports that matter
8443 / 8080	nginx (fixed dev aliases)	Yes, harmless
8000	register	Dev convenience — close it in production
8001	gateway	Dev convenience
8002	access	Dev convenience
5432	postgres	Dev convenience — close in production
9000 / 9001	minio API / console	Dev convenience
5556	dex	Only with the <code>sso</code> profile
8088	temporal-ui	Only with the <code>debug</code> profile

Hardening note. The dev ports exist because one compose file serves laptops and production. On the production VM, either bind them to `127.0.0.1` or rely on the security group to expose only 443. Nothing in the platform needs them.

Compose profiles

Profile	Brings up	When
<i>(none)</i>	The whole core stack	Always
<code>sso</code>	dex	Whenever OIDC login is required — i.e. every production run
<code>backup</code>	pgbackup (nightly <code>pg_dumpall</code> , 14-day retention)	Production
<code>debug</code>	temporal-ui	While investigating workflows

Production runs `PRISM_PROFILES="sso backup"`, which is the default in `deploy/prism-deploy.sh`.

The production posture overlay

`deploy/compose/docker-compose.prod-posture.yml` is not a different stack — it turns on the controls that default to *off* so a laptop run stays frictionless:

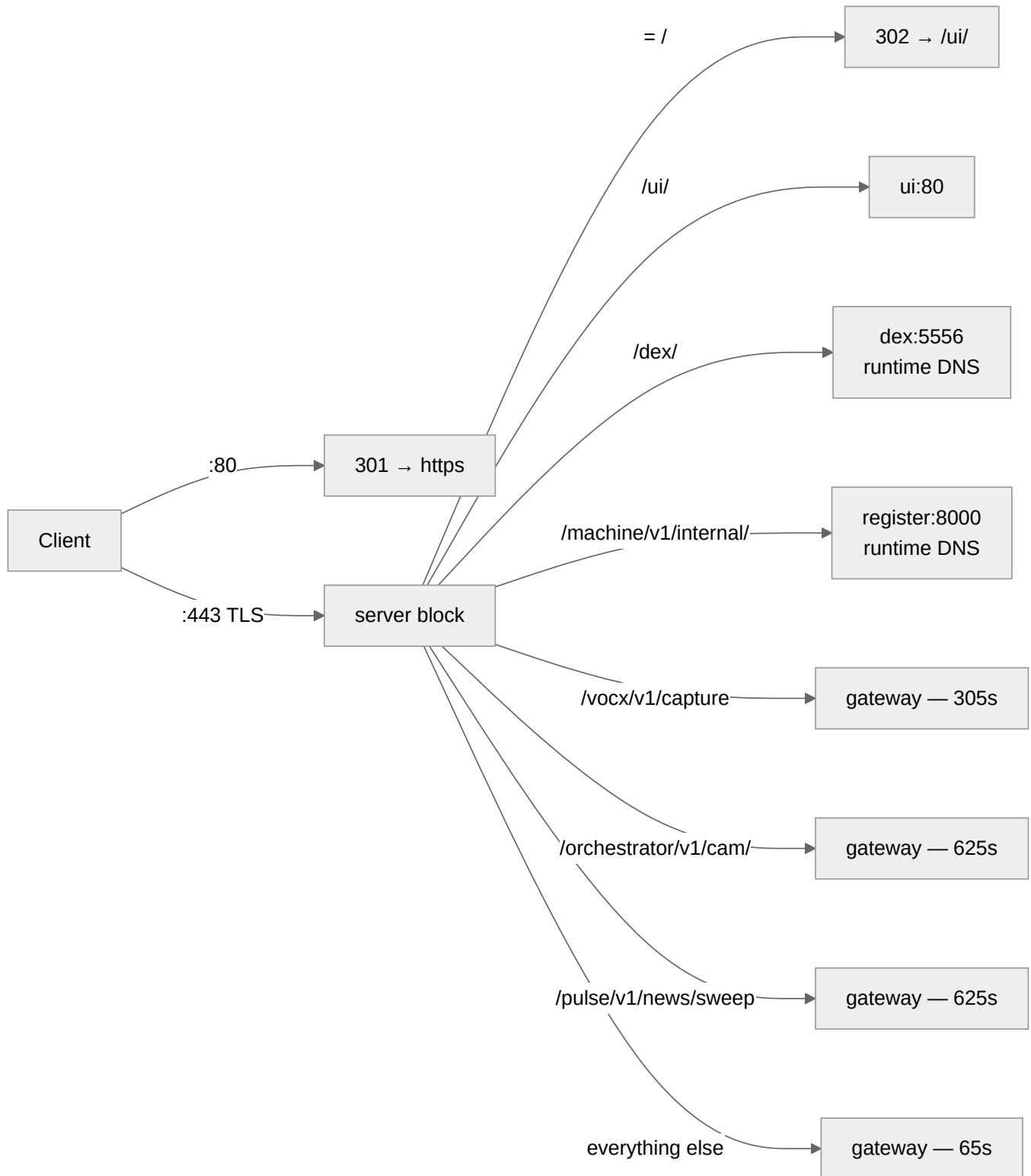
```
docker compose -f docker-compose.yml -f docker-compose.prod-posture.yml \
  --profile sso up -d --build
```

Switch	Effect
<code>GATEWAY_REQUIRE_AUTH=true</code> + issuer	Anonymous refused. <code>X-User-Email</code> is no longer trusted; identity comes only from a verified bearer.
<code>GATEWAY_OIDC_ALLOWED_DOMAINS</code>	Only <code>evamfinance.com</code> identities may authenticate. Essential once a consumer IdP such as Google is accepted — a valid Google token proves the account is real, not that it belongs to Evam.
<code>REGISTER_ENFORCE_RBAC=true</code>	An operation with no user context is refused; a bare API key can no longer act.
<code>REGISTER_ENFORCE_RLS=true</code>	PostgreSQL row-level security applied and force-converged at startup. A query cannot leave its tenant even if application code forgets a filter.
<code>REGISTER_ONLINE_REVALIDATION=true</code>	Sensitive operations (delete/restore, assignments, governed imports, evidence break-glass) revalidate against Access <i>online</i> , so a revocation takes effect immediately rather than at signed-context expiry. Fails closed (503) if Access is down.
<code>ACCESS_AUTO_SEED=false</code>	A container start never writes to a non-empty identity database; it prints a drift report instead.
<code>orchestrator_REQUIRE_AUTH</code>	An approver's identity is no longer taken on trust.

The `--profile sso` flag is required, not optional. Dex is profile-gated in the base file and an override *cannot* un-gate it — Compose filters profiled services out before merging. Without the flag you get `REQUIRE_AUTH` on with no issuer reachable, and every request 401s.

2. The edge (nginx)

`deploy/nginx/nginx.conf` is where TLS ends and where every timeout decision is made.



Timeouts are per-path, and deliberately so

Path	Read/send timeout	Why
default	65 s	A slow Register call is a fault. It should fail fast.
/vocx/v1/capture	305 s	A clip is transcribed synchronously on CPU. Cut short, the user loses a recording they already made.
/orchestrator/v1/cam/	625 s	Generating an 11-section CAM memo legitimately takes 5–10 minutes.
/pulse/v1/news/swEEP	625 s	An all-firms sweep across three sources at ~400 terms is minutes of real work.

`client_max_body_size` is **64 m** — MIS spreadsheets and scanned financials.

Two upstream styles, and the 502 they explain

```
upstream gateway { server gateway:8000; keepalive 32; } # resolved ONCE at startup
upstream ui      { server ui:80;           keepalive 8;  }

location /dex/ { resolver 127.0.0.11 valid=10s; set $dex_upstream http://dex:5556; ... }
```

A static `upstream` block resolves its DNS name **once, when nginx starts**. If `ui` or `gateway` is later recreated, Docker gives it a new IP and nginx keeps dialling the old one → **502 Bad Gateway** on a stack that is otherwise perfectly healthy.

This is a real failure that has happened in production here. Two mitigations are in place:

1. `deploy/prism-deploy.sh` calls `reload_edge()` after every swap (`nginx -s reload` , falling back to a container restart).
2. `/dex/` and `/machine/v1/internal/` use **request-time DNS** (`resolver 127.0.0.11`), so the edge starts even when those services are absent and never caches their address.

If you recreate `ui` or `gateway` by hand, run `docker compose exec nginx nginx -s reload`.

Security headers set at the edge

`X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: no-referrer`, plus a per-request `X-Request-ID` echoed back for correlation. HSTS is deliberately **not** set while the certificate is self-signed — enable it once a CA-issued pair is mounted.

Rate limiting: `limit_req zone=api` with per-path burst, and `limit_conn conn 50`.

3. Persistent state

Volume	Contents	Loss means
<code>pgdata</code>	The whole book — register, access and Temporal databases	Total data loss. Restore from <code>backups/</code> .
<code>miniodata</code>	Uploaded documents, CAM PDFs	Documents gone; database rows still reference them.
<code>pgbackups</code>	Nightly <code>pg_dumpall</code> archives, 14-day retention	Your recovery position.
<code>dexdata</code>	Dex's own storage	Re-seeded on next start.
<code>pulsedata</code>	PULSE working files	Rebuilt on next scan.
<code>deploy/vocx-secrets/</code> (bind, on disk)	Google OAuth client + per-user tokens	Users must re-authorise Google. Never in git.
<code>deploy/nginx/certs/</code> (bind, on disk)	TLS keypair	Edge will not start.
<code>deploy/compose/.env</code> (bind, on disk)	Every secret and tunable	Stack misconfigures. Never in git.

The three paths that must never enter the repository or a delivery zip: `deploy/compose/.env`, `deploy/vocx-secrets/`, `deploy/nginx/certs/`. `prism-deploy.sh` snapshots all three before an upgrade and restores them into the new tree — see [10 Upgrade & rollback](#).

One Postgres, three databases

```
postgres:16-alpine
├─ register          ← the book
├─ access            ← users, grants, matrix (created by initdb/01-create-access-db.sql)
├─ temporal          ← workflow state
└─ temporal_visibility ← workflow search
```

One server, a database per concern — matching the Helm umbrella. This is why a `pg_dumpall` is the correct backup unit: it captures all four.

4. Production filesystem layout

The deploy script expects this shape on the VM (Evam production: `/home/ubuntu/aug_11`):

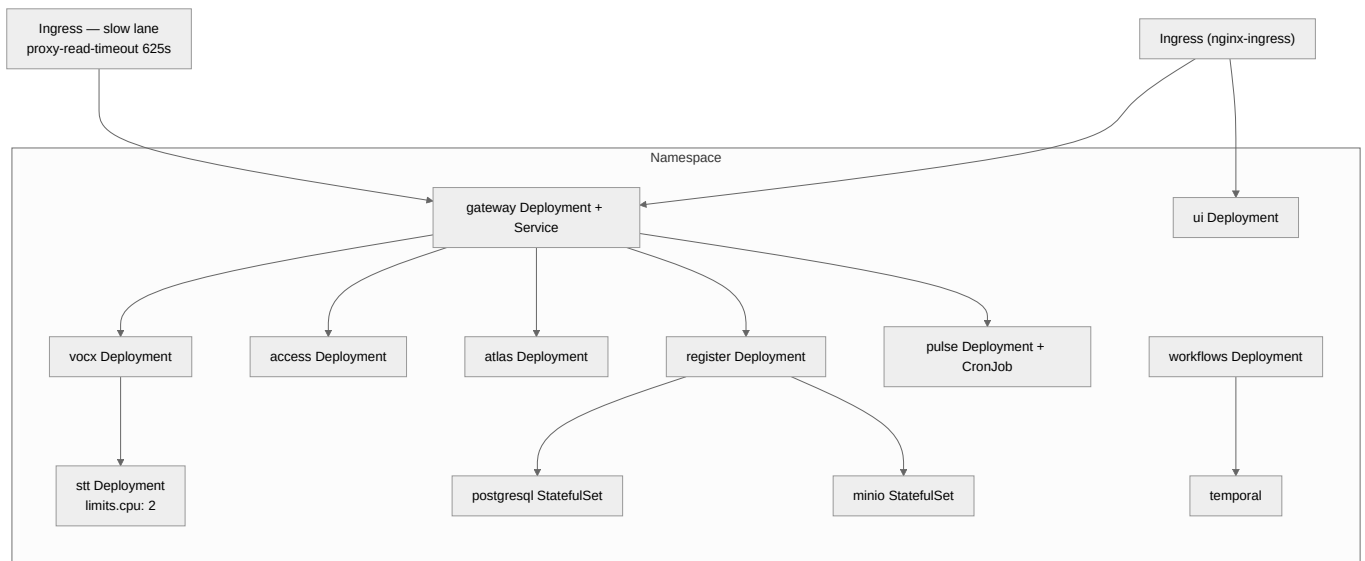
```

$PRISM_ROOT/
├─ prism/                                ← the LIVE tree compose runs from
│   └─ deploy/
│       ├── compose/.env                (secret – survives every swap)
│       ├── vocx-secrets/               (secret – survives every swap)
│       └─ nginx/certs/                 (secret – survives every swap)
├─ releases/                            ← previous trees, newest last
│   ├── prism-9bdda8b/
│   ├── prism-b423fa2/
│   └─ .previous -> prism-9bdda8b      (the rollback target; never pruned)
├─ backups/                             ← dumps + secret snapshots + deploy logs
│   ├── prism-20260819-041500.sql.gz
│   ├── secrets-20260819-041500.tar.gz
│   └─ deploy-20260819-041500.log
├─ prism-deploy.sh                      ← lives OUTSIDE the tree it swaps
└─ .prism-deploy.lock

```

The script must sit **outside** `prism/`, because a copy inside the release tree would be replaced mid-run. `PRISM_ROOT` overrides the auto-detection.

5. Helm topology



Umbrella chart at `deploy/helm/prism`, with a subchart per service: `access · atlas · gateway · minio · postgresql · pulse · register · stt · temporal · ui · vocx · workflows`.

Two Ingresses, on purpose

ingress-nginx annotations are **per-Ingress, not per-path**. A single Ingress cannot give `/vocx/v1/capture` a 625-second read timeout while keeping 65 s everywhere else. So the gateway subchart renders a second Ingress — `deploy/helm/prism/charts/gateway/templates/ingress-slow.yaml` — carrying `proxy-read-timeout` / `proxy-send-timeout` = `slowTimeoutSeconds` for the paths listed in `gateway.ingress.slowPaths` (`/vocx/v1/capture`, `/orchestrator/v1/cam`). The main Ingress carries `proxy-body-size: 64m`.

Set `slowPaths` from a **values file**, not `--set`. `--set "gateway.ingress.slowPaths=[]"` is mis-parsed by Helm's `--set` grammar; a values file behaves correctly.

The one place compose and Helm genuinely differ

	Compose	Helm
<code>stt</code> CPU limit	<code>none</code>	<code>limits.cpu: 2</code>
Correct <code>STT_CPU_THREADS</code>	derive from <code>nproc</code> on the VM	<code>2</code> (matches the pod limit)

Setting `STT_CPU_THREADS=2` on a compose box with more cores throttles transcription for no reason. See [12 VocX & STT §sizing](#).

6. Network policy and outbound access

Direction	What needs it
Inbound 443	Everything. The only port that must be open.
Outbound HTTPS	Dex → Google (if Google is the upstream IdP); PULSE → news sources; VocX → Google Drive/Docs/Calendar; notifier → SMTP.
Outbound SMTP (587/465)	The notifier, for digests and approval mails.
Internal only	Every service-to-service hop; all on the compose bridge network / cluster network.

If the deployment sits behind an egress proxy, PULSE and the VocX Google integration are the two features that will fail first, and both fail soft — the platform stays up.

7. Sizing guidance

Component	Cost driver	Notes
stt	CPU, heavily	The only genuinely CPU-hungry service. One transcription is serialised per process (ctranslate2 is not thread-safe on one model instance).
register	Modest CPU, DB connections	Scale with replicas; stateless.
postgres	RAM for cache, disk for the book + Temporal history	Temporal history grows faster than business data.
gateway / atlas / pulse / vocx	Small	Stateless, cheap to replicate.
workflows worker	Small, but must be running	A stopped worker silently stalls every in-flight process.

Practical guidance for the single-VM deployment: 4 vCPU / 16 GB is comfortable for the current desk. The binding constraint on concurrent voice captures is `nproc` versus `STT_CPU_THREADS`, covered in [12](#).

8. Quick reference — bringing the stack up

```
# Local / demo – everything, dev posture
docker compose -f deploy/compose/docker-compose.yml up --build

# Production posture (the controls on)
docker compose -f deploy/compose/docker-compose.yml \
               -f deploy/compose/docker-compose.prod-posture.yml \
               --profile sso --profile backup up -d --build

# Just the core, no workflow plane
docker compose -f deploy/compose/docker-compose.yml up postgres minio register nginx

# Helm
helm upgrade --install prism deploy/helm/prism -n prism --create-namespace -f my-values.yaml
```

For a real production upgrade, do **not** run compose by hand — use `prism-deploy.sh upgrade`, which backs up first, builds before it disturbs anything, health-gates the result and rolls itself back on failure.